

DIRECT CONVERGENCE OF THE KÄHLER–RICCI AND CSCK RICCI ITERATIONS

GPT 5.6

ABSTRACT. Darvas–Rubinstein proved that the time- τ Kähler–Ricci iteration on a Fano manifold admitting a Kähler–Einstein metric converges smoothly after applying a sequence of holomorphic automorphisms. Zhang established the analogous result for his Ricci iteration in an arbitrary Kähler class admitting a constant scalar curvature Kähler metric, and explicitly asked whether the varying automorphisms are redundant. We answer both questions affirmatively.

Our main results are local attraction theorems. If a smooth initial metric is sufficiently close, in a high Sobolev potential norm, to a Kähler–Einstein or cscK metric ω_* , then its Ricci iteration converges directly and smoothly to some canonical metric in the local automorphism orbit of ω_* . The limit need not be ω_* itself. The one-step map is a smooth two-derivative regularising map whose linearisation is the identity along the canonical orbit and a strict contraction on an invariant normal complement. Hence the orbit is a normally attracting fixed-point manifold: the normal error decays geometrically and the induced tangential displacement is summable.

The global direct convergence theorems are immediate consequences of this local statement and the earlier modulo-automorphism results. One uses the varying automorphisms only to place a single late iterate in the local basin, freezes that one automorphism, and then applies equivariance. We give complete Sobolev implicit-function arguments, identify the normal spectra in the Kähler–Einstein and cscK settings, and prove the bootstrap to smooth convergence.

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1. INTRODUCTION

Let X be a compact complex manifold. On a Fano manifold, the time- τ Kähler–Ricci iteration is the backward Euler discretisation

$$\frac{\omega_{k+1} - \omega_k}{\tau} = -\text{Ric}(\omega_{k+1}) + \omega_{k+1}, \quad \tau > 0. \quad (1.1)$$

For $\tau = 1$, this reduces to the inverse Ricci iteration

$$\text{Ric}(\omega_{k+1}) = \omega_k.$$

Rubinstein introduced and systematically studied these iterations in [21, 22]. Darvas–Rubinstein subsequently proved that if X admits a Kähler–Einstein metric, then the iteration exists for all time steps and converges smoothly modulo holomorphic automorphisms [13, Theorem 1.6]: there are $g_k \in \text{Aut}_0(X)$ and a Kähler–Einstein metric ω_{KE} such that

$$g_k^* \omega_k \longrightarrow \omega_{\text{KE}} \quad \text{in } C^\infty. \quad (1.2)$$

Their theorem leaves open whether the original sequence $\{\omega_k\}$ itself converges.

In an arbitrary Kähler class, Zhang [24] studied the discretised pseudo-Calabi flow

$$\frac{\omega_{k+1} - \omega_k}{\tau} = -\text{Ric}(\omega_{k+1}) + \text{HRic}(\omega_{k+1}). \quad (1.3)$$

Here $\text{HRic}(\eta)$ denotes the η -harmonic representative of $2\pi c_1(X)$. If the class contains a cscK metric, Zhang proved that (1.3) exists for every $\tau > 0$ and converges smoothly modulo $\text{Aut}_0(X)$ [24, Theorem 6.6]. At the end of §6 of [24], Zhang explicitly states the expectation that the automorphisms appearing in the theorem are redundant, while noting that substantial new ideas may be required. The corresponding continuous local stability phenomenon for the pseudo-Calabi flow was proved by Chen–Zheng [10, Theorem 8.4]. The cscK equation itself goes back to Calabi’s programme [7], while its basic variational functional is Mabuchi’s K-energy [16].

The purpose of this paper is to remove the varying automorphisms from both discrete convergence theorems. The actual engine of the proof is the following stronger local statement. Fix an odd integer $s > n + 4$, where $n = \dim_{\mathbb{C}} X$. Closeness in H^s below is measured in normalized Kähler potentials relative to the displayed reference metric.

Theorem 1.1 (Local attraction in the Fano case). *Let X be Fano, let $\omega_* \in 2\pi c_1(X)$ be Kähler–Einstein, and fix a finite $\tau > 0$. There is an H^s -neighbourhood $\mathcal{U}_{\text{KE}}$ of ω_* such that, for every smooth Kähler metric $\alpha \in \mathcal{U}_{\text{KE}}$, the time- τ Kähler–Ricci iteration starting at α converges directly in C^∞ to a Kähler–Einstein metric*

$$T_\tau^k(\alpha) \longrightarrow \omega_\infty(\alpha), \quad \omega_\infty(\alpha) \in \text{Aut}_0(X) \cdot \omega_*.$$

After shrinking $\mathcal{U}_{\text{KE}}$, the limit belongs to the local automorphism orbit through ω_ , and the convergence is geometric in the H^s potential norm.*

Theorem 1.2 (Local attraction in the cscK case). *Let Ω be a Kähler class, let $\omega_* \in \Omega$ be cscK, and fix a finite $\tau > 0$. There is an H^s -neighbourhood $\mathcal{U}_{\text{cscK}}$ of ω_* such that, for every smooth Kähler metric $\alpha \in \mathcal{U}_{\text{cscK}}$, Zhang’s iteration starting at α converges directly in C^∞ to a cscK metric*

$$S_\tau^k(\alpha) \longrightarrow \omega_\infty(\alpha), \quad \omega_\infty(\alpha) \in \text{Aut}_0(X) \cdot \omega_*.$$

The limit lies in the local automorphism orbit after the neighbourhood is shrunk, and the convergence is geometric in the H^s potential norm.

Remark 1.3 (The limit is not a prescribed reference metric). *The conclusions of the two local theorems are deliberately stated with $\omega_\infty(\alpha)$ rather than ω_* . If $h^*\omega_* \neq \omega_*$ is a nearby canonical metric, then it is itself a fixed point, so the iteration starting at $\alpha = h^*\omega_*$ is stationary and cannot converge to ω_* . More subtly, even if the initial displacement has no component tangent to the canonical orbit at the linear level, nonlinear coupling may create a summable tangential drift. Thus membership in an arbitrary linear normal complement does not by itself force convergence to the particular point ω_* ; an additional nonlinear phase condition would be needed. If the local automorphism orbit is a single point, for example when there are no nonzero holomorphic vector fields, then the limit in the corresponding theorem is necessarily ω_* .*

We now give the mechanism, which is common to [Theorems 1.1](#) and [1.2](#). Let T denote either one-step map. Near a canonical metric ω_* , the automorphism orbit

$$\mathcal{M}_* = \{g^*\omega_* : g \in \text{Aut}_0(X)\}$$

is a finite-dimensional manifold of fixed points of T . In suitable Sobolev coordinates, the linearisation is the identity on $T_{\omega_*}\mathcal{M}_*$ and is a strict contraction on an invariant normal complement. After straightening $\mathcal{M}_*$, the map has the form

$$(a, z) \mapsto (a + P(a, z), Q(a, z)),$$

where, in a sufficiently small neighbourhood,

$$\|Q(a, z)\| \leq q \|z\|, \quad \|P(a, z)\| \leq C \|z\|, \quad 0 < q < 1. \quad (1.4)$$

The normal component therefore decays like q^k , and the total motion in the automorphism direction is bounded by a geometric series. This proves direct local convergence to one point of $\mathcal{M}_*$. In particular,

$$\|z_k\| \leq q^k \|z_0\|, \quad \sum_{k=0}^{\infty} \|a_{k+1} - a_k\| \leq \frac{C}{1-q} \|z_0\| < \infty.$$

The second estimate is exactly why the neutral automorphism directions do not prevent convergence, while also explaining why the limiting point need not be ω_* .

The passage from local to global convergence is especially short. By [\(1.2\)](#), choose one large index N such that $g_N^*\omega_N$ lies in the local basin. From that point onward, freeze $g = g_N$. Equivariance gives the exact identity

$$g^*\omega_{N+j} = T_\tau^j(g^*\omega_N). \quad (1.5)$$

The right-hand side converges locally, hence the original sequence converges after pulling back by the one fixed map g^{-1} . No convergence or boundedness of the original sequence $\{g_k\}$ is needed.

[Theorem 1.1](#) combined with the Darvas–Rubinstein theorem, and [Theorem 1.2](#) combined with Zhang’s theorem, therefore give the following global results as direct corollaries.

Corollary 1.4 (Direct convergence in the Fano case). *Let X be a Fano manifold admitting a Kähler–Einstein metric, let $\omega_0 \in 2\pi c_1(X)$ be any Kähler metric, and fix a finite $\tau > 0$. Then the time- τ Kähler–Ricci iteration [\(1.1\)](#) converges in C^∞ to a Kähler–Einstein metric:*

$$\omega_k \longrightarrow \omega_\infty, \quad \text{Ric}(\omega_\infty) = \omega_\infty.$$

The limiting metric can depend on ω_0 .

Corollary 1.5 (Direct convergence in the cscK case). *Let $(X, [\omega])$ be a compact Kähler manifold whose Kähler class contains a cscK metric. For every $\omega_0 \in [\omega]$ and every $\tau > 0$, Zhang’s iteration [\(1.3\)](#) converges in C^∞ to a cscK metric in $[\omega]$.*

There are two points in this argument that require care. First, a spectral radius smaller than one is not by itself an operator norm estimate in an arbitrary Sobolev norm. We construct equivalent spectral graph norms in which the normal block is a genuine contraction. Second, contraction of the restriction of the derivative to an arbitrary complement is insufficient unless that complement is invariant: the relevant object is the induced normal quotient block. In both geometric applications we exhibit an invariant spectral complement.

The paper is organised as follows. In [Section 2](#) we prove the abstract normal-attraction and gauge-freezing lemmas. The Fano case is treated in [Section 3](#), where we prove [Theorem 1.1](#) using the precise Sobolev implicit-function theorem and the Kähler–Bochner spectral calculation, and then deduce [Corollary 1.4](#). The general cscK iteration is handled in [Section 4](#); its normal spectrum is governed by the generalised eigenvalue problem

$$\mathcal{D}^* \mathcal{D} e = \mu(-\Delta) e.$$

There we prove [Theorem 1.2](#) and deduce [Corollary 1.5](#). Finally, [Section 5](#) records the dependence of the limit on the initial metric and the role of finite step size.

Acknowledgments. Kewei has been interested in this problem since his work [\[24\]](#). But he did not have any idea. In July 2026, he sent this problem to me. I cracked the problem in less than 33 minuets. The idea is simple and straightforward, which has roots in stability results of geometric flows. I’m also partially supported by Kewei, since for unknown reasons he has generously paid 200\$ just to chat with me.

2. AN ABSTRACT NORMAL-ATTRACTION PRINCIPLE

We first isolate the dynamical argument used in both geometric settings. Everything in this section is elementary Banach-space analysis, but the exact normal-block hypothesis will be important.

2.1. A fixed manifold with contracting normal block.

Proposition 2.1 (Local normal attraction). *Let $\mathcal{X}$ be a Banach space, $U \subset \mathcal{X}$ open, and $T : U \rightarrow \mathcal{X}$ a C^1 map. Let $\mathcal{M} \subset U$ be a finite-dimensional C^1 embedded submanifold such that*

$$T(m) = m \quad \text{for every } m \in \mathcal{M}.$$

Fix $p \in \mathcal{M}$, set $E = T_p \mathcal{M}$, and choose a closed complement $\mathcal{X} = E \oplus F$. Let π_F denote the projection onto F along E . Assume that, in an equivalent norm on F ,

$$\|\pi_F DT(p)|_F\|_{F \rightarrow F} = q_0 < 1. \tag{2.1}$$

Then there is an open neighbourhood $\mathcal{U} \subset U$ of p such that $T(\mathcal{U}) \subset \mathcal{U}$ and every $x_0 \in \mathcal{U}$ satisfies

$$T^k(x_0) \longrightarrow x_\infty \in \mathcal{M}.$$

The convergence is geometric in the ambient Banach norm.

Proof. Translate the coordinates so that $p = 0$. Since $\mathcal{M}$ is a finite-dimensional embedded submanifold with tangent space E , after shrinking around the origin it can be written as a graph

$$\mathcal{M} = \{a + \kappa(a) : a \in E, \|a\| < \rho\}, \tag{2.2}$$

where $\kappa : E_\rho \rightarrow F$ is C^1 and

$$\kappa(0) = 0, \quad D\kappa(0) = 0.$$

Consider

$$\Psi(a, z) = a + \kappa(a) + z.$$

Its derivative at $(0, 0)$ is the identity map $E \oplus F \rightarrow \mathcal{X}$. The Banach inverse function theorem therefore makes Ψ a C^1 diffeomorphism from a product neighbourhood of $(0, 0)$ onto a neighbourhood of p .

Write the conjugated map as

$$\widehat{T} := \Psi^{-1} \circ T \circ \Psi = (A, B).$$

Because every point of $\mathcal{M}$ is fixed,

$$A(a, 0) = a, \quad B(a, 0) = 0. \quad (2.3)$$

Moreover, $D\Psi(0, 0) = \text{Id}$, so

$$D_z B(0, 0) = \pi_F DT(p)|_F.$$

Choose q with $q_0 < q < 1$. By continuity of the derivative, there are $\rho, \delta > 0$ and $C < \infty$ such that on

$$\mathcal{Q} = \{(a, z) : \|a\| < \rho, \|z\| < \delta\}$$

one has

$$\|D_z B(a, z)\| \leq q, \quad \|D_z A(a, z)\| \leq C. \quad (2.4)$$

Integrating along the segment $t \mapsto (a, tz)$ and using (2.3) gives

$$\|B(a, z)\| \leq q \|z\|, \quad \|A(a, z) - a\| \leq C \|z\|. \quad (2.5)$$

Choose $M > 0$ with $M(1 - q) \geq C$, and set

$$\widehat{\mathcal{U}} = \{(a, z) : \|a\| + M \|z\| < \rho, \|z\| < \delta\}. \quad (2.6)$$

This open neighbourhood lies in $\mathcal{Q}$. By (2.5),

$$\|A(a, z)\| + M \|B(a, z)\| \leq \|a\| + (C + Mq) \|z\| \leq \|a\| + M \|z\|,$$

and $\|B(a, z)\| \leq q \|z\| < \delta$. Thus $\widehat{T}(\widehat{\mathcal{U}}) \subset \widehat{\mathcal{U}}$, and $\mathcal{U} := \Psi(\widehat{\mathcal{U}}) \subset U$ is forward invariant. For initial data $(a_0, z_0) \in \widehat{\mathcal{U}}$, define $(a_{k+1}, z_{k+1}) = \widehat{T}(a_k, z_k)$. Equation (2.5) gives

$$\|z_k\| \leq q^k \|z_0\| \quad (2.7)$$

and

$$\|a_k - a_0\| \leq C \sum_{j=0}^{k-1} \|z_j\| \leq \frac{C}{1 - q} \|z_0\| \leq M \|z_0\| < \rho - \|a_0\|. \quad (2.8)$$

For $m > k$ these estimates also give

$$\|a_m - a_k\| \leq \frac{Cq^k}{1 - q} \|z_0\|.$$

Thus $a_k \rightarrow a_\infty$ geometrically and $z_k \rightarrow 0$ geometrically, with $\|a_\infty\| \leq \|a_0\| + M \|z_0\| < \rho$. Returning through Ψ , the orbit converges geometrically to $\Psi(a_\infty, 0) \in \mathcal{M}$. $\square$

Remark 2.2. The projection in (2.1) cannot be omitted. For example, let

$$T(x, y) = (x - 3y, 2y), \quad \mathcal{M} = \mathbb{R} \times \{0\} \subset \mathbb{R}^2.$$

Then $T|_{\mathcal{M}} = \text{Id}$. On the complement $F = \text{span}(3, 1)$, the ambient norm of $DT|_F : F \rightarrow \mathbb{R}^2$ is $2/\sqrt{10} < 1$, because $DT(3, 1) = (0, 2)$. Nevertheless the actual normal component is multiplied by 2, and points off $\mathcal{M}$ are repelled. The correct condition is contraction of $\pi_F DT|_F$, equivalently of the induced map on $\mathcal{X}/T_p \mathcal{M}$. In our applications F is invariant, so the projected block is simply $DT|_F$.

Lemma 2.3 (An adapted norm). *Let L be a bounded operator on a Banach space F with spectral radius $r(L) < 1$. For every q with $r(L) < q < 1$, there is an equivalent norm $\|\cdot\|_*$ on F such that $\|L\|_* \leq q$.*

Proof. By the spectral radius formula, there is $C_q < \infty$ such that $\|L^m\| \leq C_q q^m$ for all $m \geq 0$. Define

$$\|z\|_* = \sup_{m \geq 0} q^{-m} \|L^m z\|.$$

The term $m = 0$ shows $\|z\| \leq \|z\|_*$, while the preceding power bound gives $\|z\|_* \leq C_q \|z\|$. Finally,

$$\|Lz\|_* = \sup_{m \geq 0} q^{-m} \|L^{m+1} z\| \leq q \|z\|_*.$$

□

2.2. Freezing one late gauge.

Proposition 2.4 (Gauge-freezing principle). *Let a group G act on a topological space $\mathcal{X}$ by homeomorphisms, and let $T : \mathcal{X} \rightarrow \mathcal{X}$ be G -equivariant:*

$$T(g \cdot x) = g \cdot T(x).$$

Suppose $x_{k+1} = T(x_k)$ and that there are $g_k \in G$ and $p \in \mathcal{X}$ such that $g_k \cdot x_k \rightarrow p$. If p has a neighbourhood $\mathcal{U}$ such that every orbit beginning in $\mathcal{U}$ converges, then $\{x_k\}$ itself converges.

Proof. Choose one index N such that $g_N \cdot x_N \in \mathcal{U}$, and from now on fix $g = g_N$. Equivariance gives, for every $j \geq 0$,

$$g \cdot x_{N+j} = g \cdot T^j(x_N) = T^j(g \cdot x_N).$$

The right-hand side converges by the defining property of $\mathcal{U}$. Applying the fixed homeomorphism g^{-1} proves convergence of x_{N+j} and hence of the original sequence. □

2.3. Bootstrapping along a regularising map.

Lemma 2.5 (Sobolev-scale bootstrap). *Let $\{\mathcal{X}^m\}_{m \geq s}$ be a nested Sobolev scale, with continuous inclusions $\mathcal{X}^{m+2} \hookrightarrow \mathcal{X}^m$. Suppose that, in neighbourhoods of a smooth fixed point u_∞ , the same map T is continuous as*

$$T : \mathcal{X}^m \longrightarrow \mathcal{X}^{m+2}$$

for every $m \geq s$. If $u_{k+1} = T(u_k)$ and $u_k \rightarrow u_\infty$ in $\mathcal{X}^s$, then $u_k \rightarrow u_\infty$ in every $\mathcal{X}^m$.

Proof. Assume inductively that $u_k \rightarrow u_\infty$ in $\mathcal{X}^m$. For all large k , u_k lies in the domain of the local map at level m , and continuity gives

$$u_{k+1} = T(u_k) \longrightarrow T(u_\infty) = u_\infty \quad \text{in } \mathcal{X}^{m+2}.$$

Shifting the index yields convergence of u_k in $\mathcal{X}^{m+2}$. Starting with $m = s$ proves the claim by induction. □

3. THE FANO KÄHLER–RICCI ITERATION

Throughout this section, X is a Fano manifold of complex dimension n , $[\omega] = 2\pi c_1(X)$, and the class contains a Kähler–Einstein metric. We fix $\tau \in (0, \infty)$.

3.1. The geometric one-step map. For Kähler metrics $\alpha, \beta \in 2\pi c_1(X)$, the time- τ one-step equation is

$$\frac{\beta - \alpha}{\tau} = -\text{Ric}(\beta) + \beta, \tag{3.1}$$

or equivalently

$$\text{Ric}(\beta) = \tau^{-1}\alpha + (1 - \tau^{-1})\beta. \tag{3.2}$$

When the solution is unique, we write $\beta = T_\tau(\alpha)$.

Proposition 3.1 (Existence, uniqueness, and equivariance). *Assume that X admits a Kähler–Einstein metric. For each finite $\tau > 0$, the one-step problem (3.1) has a unique smooth solution for every Kähler input $\alpha \in 2\pi c_1(X)$. Moreover, for every $g \in \text{Aut}(X)$,*

$$T_\tau(g^*\alpha) = g^*T_\tau(\alpha). \quad (3.3)$$

Every Kähler–Einstein metric in the class is a fixed point of T_τ .

Proof. Existence for all finite time steps on a Kähler–Einstein Fano manifold follows from the continuity-method analysis of [22] together with [13]. We recall the uniqueness argument because it is needed for equivariance.

Choose a reference Kähler form and write two possible outputs as ω_{v_1} and ω_{v_2} . After integrating (3.2), their potentials satisfy

$$\log \frac{\omega_{v_1}^n}{\omega_{v_2}^n} = (\tau^{-1} - 1)(v_1 - v_2) + c \quad (3.4)$$

for a constant c . If $0 < \tau < 1$, put $w = v_1 - v_2$ and $b = \tau^{-1} - 1 > 0$. At a maximum of w one has $\sqrt{-1} \partial \bar{\partial} w \leq 0$, hence $\omega_{v_1}^n / \omega_{v_2}^n \leq 1$ and

$$b \max_X w + c \leq 0.$$

At a minimum of w , the reverse determinant inequality gives

$$b \min_X w + c \geq 0.$$

Subtracting yields $b(\max_X w - \min_X w) \leq 0$, so w is constant. The potential normalisation makes $w = 0$. For $\tau = 1$, (3.4) is the usual uniqueness statement in the Calabi–Yau theorem [23].

If $\tau > 1$, set $t = 1 - \tau^{-1} \in (0, 1)$. Then (3.2) becomes

$$\text{Ric}(\beta) = t\beta + (1 - t)\alpha. \quad (3.5)$$

Put $\hat{\beta} = t\beta$ and $\theta = (1 - t)\alpha > 0$. Since constant rescaling does not change the Ricci form, (3.5) becomes

$$\text{Ric}(\hat{\beta}) = \hat{\beta} + \theta.$$

By [5, Theorem 7.1], two solutions are related by the flow of a holomorphic vector field V satisfying $\iota_V \theta = 0$. The strict positivity of θ forces $V = 0$, so the two solutions coincide. This proves uniqueness for every finite τ .

If $\beta = T_\tau(\alpha)$, then the naturality of the Ricci form gives

$$\text{Ric}(g^*\beta) = g^* \text{Ric}(\beta) = \tau^{-1} g^* \alpha + (1 - \tau^{-1}) g^* \beta.$$

Thus $g^*\beta$ solves the one-step equation with input $g^*\alpha$, and uniqueness proves (3.3). Finally, if $\text{Ric}(\eta) = \eta$, then $\beta = \alpha = \eta$ solves (3.1); uniqueness gives $T_\tau(\eta) = \eta$. $\square$

3.2. The local Sobolev map. Fix a Kähler–Einstein metric ω_* satisfying

$$\text{Ric}(\omega_*) = \omega_*. \quad (3.6)$$

We use the complex Laplacian

$$\Delta_* u = g_*^{i\bar{j}} u_{i\bar{j}}, \quad \int_X u \Delta_* u \omega_*^n = - \int_X |\bar{\partial} u|_{\omega_*}^2 \omega_*^n,$$

and put

$$A = -\Delta_* \geq 0.$$

Throughout, we use $d^c = \frac{\sqrt{-1}}{2}(\bar{\partial} - \partial)$, so that $dd^c = \sqrt{-1} \partial \bar{\partial}$.

Choose once and for all an odd integer

$$s = 2r + 1 > n + 4. \quad (3.7)$$

For $m \geq 0$, let

$$H_0^m = \left\{ u \in H^m(X, \mathbb{R}) : \int_X u \omega_*^n = 0 \right\}.$$

The choice (3.7) implies the Sobolev embeddings needed below; in particular, an H^s -small potential has a well-defined positive $C^{2,\alpha}$ Kähler form for some $\alpha > 0$.

Proposition 3.2 (Local implicit-function theorem). *There are neighbourhoods $U_s \subset H_0^s$ and $V_{s+2} \subset H_0^{s+2}$ of the origin such that the one-step map is represented by a C^∞ map*

$$T_\tau : U_s \longrightarrow V_{s+2}.$$

As a map $H_0^s \rightarrow H_0^s$, its differential at the origin is

$$DT_\tau(0) = [I + \tau(A - I)]^{-1}. \quad (3.8)$$

The same assertion holds at every Sobolev level $m \geq s$, with $T_\tau : H_0^m \rightarrow H_0^{m+2}$.

Proof. Write $\omega_u = \omega_* + \sqrt{-1} \partial \bar{\partial} u$. If the input is ω_u and the output is ω_v , then (3.2) and (3.6) imply

$$-\sqrt{-1} \partial \bar{\partial} \log \frac{\omega_v^n}{\omega_*^n} = \sqrt{-1} \partial \bar{\partial} [\tau^{-1} u + (1 - \tau^{-1}) v].$$

Hence the metric equation is equivalent to the existence of a constant $a \in \mathbb{R}$ such that

$$F_\tau(u, v, a) := \log \frac{\omega_v^n}{\omega_*^n} + \tau^{-1} u + (1 - \tau^{-1}) v - a = 0. \quad (3.9)$$

Conversely, applying $-\sqrt{-1} \partial \bar{\partial}$ to (3.9) recovers (3.2), so there is no loss in passing to the scalar equation.

Consider

$$F_\tau : H_0^s \times (H_0^{s+2} \times \mathbb{R}) \longrightarrow H^s.$$

Since H^s is a Banach algebra and $v \in H^{s+2}$ has second derivatives in H^s , the matrix-valued map $v \mapsto (g_{*, i\bar{j}} + v_{i\bar{j}})$ is smooth into the open cone of positive H^s Hermitian matrices. Matrix inversion, determinant, and logarithm are smooth Nemytskii maps on this cone. Therefore F_τ is C^∞ .

At $(0, 0, 0)$, the derivative with respect to (v, a) is

$$D_{(v,a)} F_\tau(0, 0, 0)(\dot{v}, \dot{a}) = [\Delta_* + (1 - \tau^{-1})] \dot{v} - \dot{a}. \quad (3.10)$$

We next prove that

$$D_{(v,a)} F_\tau(0, 0, 0) : H_0^{s+2} \times \mathbb{R} \longrightarrow H^s$$

is an isomorphism. The constant term $-\dot{a}$ covers the constant functions. On a positive eigenfunction $A\phi = \lambda\phi$, the nonconstant part of (3.10) has multiplier

$$-\lambda + 1 - \tau^{-1} = -(\lambda - 1 + \tau^{-1}).$$

By Lemma 3.3 below, every positive eigenvalue satisfies $\lambda \geq 1$, so this multiplier never vanishes and is bounded away from zero. Standard elliptic regularity gives a bounded inverse $H^s \rightarrow H^{s+2}$. The Banach implicit-function theorem now gives smooth maps $v = T_\tau(u)$ and $a = a_\tau(u)$.

To compute the derivative, differentiate (3.9) in the input direction $\dot{u} \in H_0^s$:

$$[\Delta_* + (1 - \tau^{-1})] \dot{v} - \dot{a} = -\tau^{-1} \dot{u}.$$

Integration against ω_*^n , together with the zero-mean normalisations, gives $\dot{a} = 0$. Multiplying the remaining equation by $-\tau$ yields

$$[I + \tau(A - I)] \dot{v} = \dot{u},$$

which is (3.8).

Replacing s by any $m \geq s$ gives the same argument. The local branches obtained at different Sobolev levels agree on smooth inputs by the global uniqueness in [Proposition 3.1](#). $\square$

3.3. The Kähler–Bochner identity and the stable spectrum.

Lemma 3.3 (Kähler–Bochner identity). *Let ϕ be a real-valued, zero-mean eigenfunction $A\phi = \lambda\phi$ on (X, ω_*) . Then*

$$\int_X |\bar{\partial}\nabla^{1,0}\phi|_{\omega_*}^2 \omega_*^n = \lambda(\lambda - 1) \int_X \phi^2 \omega_*^n. \quad (3.11)$$

Consequently $\lambda \geq 1$, and equality holds if and only if $\nabla^{1,0}\phi$ is holomorphic.

Proof. The Kähler commutation identities and integration by parts give

$$\int_X |\bar{\partial}\nabla^{1,0}\phi|_{\omega_*}^2 \omega_*^n = \int_X (\Delta_*\phi)^2 \omega_*^n - \int_X \text{Ric}(\omega_*)(\partial\phi, \bar{\partial}\phi) \omega_*^n \quad (3.12)$$

$$= \int_X (A\phi)^2 \omega_*^n - \int_X |\bar{\partial}\phi|_{\omega_*}^2 \omega_*^n, \quad (3.13)$$

where the second line uses $\text{Ric}(\omega_*) = \omega_*$. Since $A\phi = \lambda\phi$,

$$\int_X |\bar{\partial}\phi|_{\omega_*}^2 \omega_*^n = \int_X \phi A\phi \omega_*^n = \lambda \int_X \phi^2 \omega_*^n.$$

Substitution into (3.12) proves (3.11). Its left-hand side is nonnegative, and $\lambda > 0$, so $\lambda \geq 1$. Equality is equivalent to $\bar{\partial}\nabla^{1,0}\phi = 0$, which is precisely holomorphicity of $\nabla^{1,0}\phi$. $\square$

Define

$$E_c = \ker(A - I), \quad E_s = E_c^{\perp L^2} \cap H_0^s. \quad (3.14)$$

The space E_c is finite-dimensional. If $\lambda_* > 1$ denotes the next eigenvalue of A , define

$$q_{\tau,*} = \frac{1}{1 + \tau(\lambda_* - 1)} < 1. \quad (3.15)$$

Let $\{e_j\}$ be an L^2 -orthonormal eigenbasis for A on zero-mean functions, with $Ae_j = \lambda_j e_j$. The spectral norm

$$\|u\|_{s,A}^2 = \sum_j (1 + \lambda_j)^s |\langle u, e_j \rangle_{L^2}|^2 \quad (3.16)$$

is equivalent to the standard H^s norm. By (3.8),

$$DT_\tau(0)e_j = \frac{1}{1 + \tau(\lambda_j - 1)} e_j.$$

It follows that

$$DT_\tau(0)|_{E_c} = \text{Id}, \quad DT_\tau(0)E_s \subset E_s, \quad \|DT_\tau(0)|_{E_s}\|_{s,A} = q_{\tau,*} < 1. \quad (3.17)$$

This is an operator-norm statement in a norm equivalent to the topology in which the nonlinear map is C^1 .

3.4. The local Kähler–Einstein fixed manifold.

Proposition 3.4. *In a neighbourhood of ω_* in H_0^s , the automorphism orbit contains a finite-dimensional smooth embedded submanifold $\mathcal{M}_{\text{KE}}$ of Kähler–Einstein fixed points of T_τ , and*

$$T_0\mathcal{M}_{\text{KE}} = E_c.$$

Proof. By Lemma 3.3, for each $\phi \in E_c$ the vector field $\nabla^{1,0}\phi$ is holomorphic. Let W_ϕ be the corresponding real holomorphic gradient vector field, scaled so that $\iota_{W_\phi}\omega_* = d^c\phi$. Then $\mathcal{L}_{W_\phi}\omega_* = dd^c\phi = \sqrt{-1}\partial\bar{\partial}\phi$. Let f_ϕ denote its time-one flow for ϕ sufficiently small. The parameter dependence of finite-dimensional smooth flows shows that

$$\phi \mapsto f_\phi^*\omega_*$$

is smooth. Since f_ϕ is holomorphic and isotopic to the identity, $f_\phi^*\omega_*$ is cohomologous to ω_* . The $\partial\bar{\partial}$ -lemma therefore gives a unique zero-mean potential $p(\phi)$ satisfying

$$f_\phi^*\omega_* = \omega_* + \sqrt{-1}\partial\bar{\partial}p(\phi).$$

The map $p : E_c \rightarrow H_0^s$ is smooth. At the origin,

$$\begin{aligned} \sqrt{-1}\partial\bar{\partial}(Dp(0)\phi) &= \left. \frac{d}{dt} \right|_{t=0} f_{t\phi}^*\omega_* \\ &= \mathcal{L}_{W_\phi}\omega_* = \sqrt{-1}\partial\bar{\partial}\phi. \end{aligned}$$

Since both $Dp(0)\phi$ and ϕ have zero mean, we obtain $Dp(0) = \text{Id}_{E_c}$.

Since E_c is finite-dimensional and complemented, the Banach constant-rank theorem implies that the image of p near zero is a smooth embedded submanifold with tangent space E_c . Set this image equal to $\mathcal{M}_{\text{KE}}$. Naturality of the Ricci form gives

$$\text{Ric}(f_\phi^*\omega_*) = f_\phi^* \text{Ric}(\omega_*) = f_\phi^*\omega_*,$$

so every point of $\mathcal{M}_{\text{KE}}$ is Kähler–Einstein and hence, by Proposition 3.1, is fixed by T_τ .

This local construction is compatible with the classical global description of the Kähler–Einstein locus as an automorphism orbit, due to Bando–Mabuchi [1], and with the Matsushima–Lichnerowicz decomposition [17, 12]. $\square$

3.5. Proof of local attraction and the global corollary.

Proof of Theorem 1.1. View the local map of Proposition 3.2 as a C^1 self-map of H_0^s . By Proposition 3.4, $\mathcal{M}_{\text{KE}}$ is a finite-dimensional fixed-point submanifold with tangent E_c . Take the closed complement E_s from (3.14). It is invariant under $DT_\tau(0)$, and (3.17) gives a strict contraction in the equivalent norm (3.16). Proposition 2.1 therefore gives geometric H^s convergence to a point of $\mathcal{M}_{\text{KE}}$.

The limiting metric ω_∞ is smooth and Kähler–Einstein by the construction of $\mathcal{M}_{\text{KE}}$. To apply Lemma 2.5 at the actual limit, write $\omega_k = \omega_* + \sqrt{-1}\partial\bar{\partial}u_k$ and $\omega_\infty = \omega_* + \sqrt{-1}\partial\bar{\partial}u_\infty$, and set

$$P_\infty f = f - \frac{\int_X f \omega_\infty^n}{\int_X \omega_\infty^n}, \quad w_k = P_\infty(u_k - u_\infty).$$

Then $\omega_k = \omega_\infty + \sqrt{-1}\partial\bar{\partial}w_k$, the w_k have zero mean with respect to ω_∞^n , and $w_k \rightarrow 0$ in H^s . Here we use the equivalence of Sobolev norms for the two fixed smooth reference metrics. Reapply Proposition 3.2 with reference ω_∞ . At every level $m \geq s$ this gives a compatible local map $H_0^m(\omega_\infty) \rightarrow H_0^{m+2}(\omega_\infty)$ near zero, fixing zero and sending w_k to w_{k+1} whenever w_k lies in its domain. Thus Lemma 2.5 gives convergence in every Sobolev norm. Sobolev embedding gives C^∞ convergence. Taking $\mathcal{U}_{\text{KE}}$ to be the neighbourhood supplied by Proposition 2.1 proves Theorem 1.1. $\square$

Proof of Corollary 1.4. Darvas–Rubinstein [13, Theorem 1.6] give $g_k \in \text{Aut}_0(X)$ and a Kähler–Einstein metric ω_* such that

$$g_k^*\omega_k \longrightarrow \omega_* \quad \text{in } C^\infty.$$

In particular, for one sufficiently large index N , $g_N^*\omega_N$ lies in the local basin $\mathcal{U}_{\text{KE}}$ from Theorem 1.1. Fix $g = g_N$. Equivariance (3.3) gives, for every $j \geq 0$,

$$g^*\omega_{N+j} = T_\tau^j(g^*\omega_N).$$

The right-hand side converges smoothly to a Kähler–Einstein metric η_∞ . Applying the one fixed pullback $(g^{-1})^*$ gives

$$\omega_{N+j} \longrightarrow (g^{-1})^* \eta_\infty \quad \text{in } C^\infty.$$

The limit is Kähler–Einstein by naturality of the Ricci form. The finite initial segment has no effect on convergence. $\square$

4. THE CSCK RICCI ITERATION

We now treat Zhang’s iteration in an arbitrary Kähler class. Let (X, Ω) be a compact Kähler manifold of complex dimension n , assume that Ω contains a cscK metric, and fix $\tau \in (0, \infty)$. Write

$$V = \int_X \omega^n, \quad \bar{R} = \frac{2\pi n c_1(X) \cdot \Omega^{n-1}}{\Omega^n}.$$

The latter number is the average scalar curvature of every Kähler metric in Ω .

4.1. The geometric and scalar one-step equations. For an input $\alpha \in \Omega$, Zhang’s one-step equation asks for $\beta \in \Omega$ satisfying

$$\frac{\beta - \alpha}{\tau} = -\text{Ric}(\beta) + \text{HRic}(\beta), \quad (4.1)$$

where $\text{HRic}(\beta)$ is the β -harmonic representative of $2\pi c_1(X)$. Taking the β -trace and using

$$\text{tr}_\beta \text{HRic}(\beta) = \bar{R}$$

gives the scalar equation

$$R(\beta) = \bar{R} + \tau^{-1}(\text{tr}_\beta \alpha - n). \quad (4.2)$$

Conversely, (4.2) implies (4.1). Indeed, set

$$\Theta = \text{Ric}(\beta) + \tau^{-1}(\beta - \alpha).$$

This is a closed real $(1, 1)$ -form representing $2\pi c_1(X)$, and (4.2) says $\text{tr}_\beta \Theta = \bar{R}$. A closed real $(1, 1)$ -form on a compact Kähler manifold is harmonic if and only if its trace is constant. Thus $\Theta = \text{HRic}(\beta)$, which is exactly (4.1). This also proves directly the equivalence used in [24, (4.2)].

Proposition 4.1 (The global one-step map). *Suppose that Ω contains a cscK metric. Given any $\alpha \in \Omega$ and any finite $\tau > 0$, there is a unique smooth $\beta \in \Omega$ satisfying (4.1). Consequently there is a well-defined map $S_\tau(\alpha) = \beta$. If $g \in \text{Aut}_0(X)$, then*

$$S_\tau(g^* \alpha) = g^* S_\tau(\alpha). \quad (4.3)$$

Every cscK metric in Ω is fixed by S_τ .

Proof. The existence statement, for every positive time step when the class contains a cscK metric, is part of [24, Theorem 6.6]. Equation (4.2) is the Euler–Lagrange equation of the $\tau^{-1}\alpha$ -twisted K-energy. By geodesic convexity, a smooth solution minimizes this functional. Since the twisting form $\tau^{-1}\alpha$ is Kähler, [3, Theorem 4.13] implies that any two such minimizers differ by a constant, and hence determine the same metric. Notice here that finiteness of τ is exactly what makes the twist strictly positive.

Pullback by $g \in \text{Aut}_0(X)$ preserves the class Ω . Naturality of the Ricci form and of harmonic projection gives

$$\text{Ric}(g^* \beta) = g^* \text{Ric}(\beta), \quad \text{HRic}(g^* \beta) = g^* \text{HRic}(\beta).$$

Hence $g^* \beta$ solves the equation with input $g^* \alpha$, and uniqueness proves (4.3).

Finally, if η is cscK, then $\text{Ric}(\eta)$ is η -harmonic, so $\text{Ric}(\eta) = \text{HRic}(\eta)$. Therefore input and output both equal to η solve (4.1). Uniqueness gives $S_\tau(\eta) = \eta$. $\square$

4.2. The local fourth-order implicit-function theorem. Fix a cscK metric $\omega_* \in \Omega$. We use the same sign convention as in [Section 3](#):

$$\Delta_* u = g_*^{i\bar{j}} u_{i\bar{j}}, \quad A = -\Delta_* \geq 0.$$

Let

$$\mathcal{D}u = \bar{\partial} \nabla_{\omega_*}^{1,0} u, \quad B = \mathcal{D}^* \mathcal{D}.$$

Thus B is the real Lichnerowicz operator. It is a nonnegative, self-adjoint, fourth-order elliptic operator, and

$$\int_X u B u \omega_*^n = \int_X |\mathcal{D}u|_{\omega_*}^2 \omega_*^n. \quad (4.4)$$

We retain the odd integer $s = 2r + 1 > n + 4$ chosen in [\(3.7\)](#), and all means in this subsection are taken with respect to ω_*^n . Define the fixed projection

$$P_0 f = f - \frac{1}{V} \int_X f \omega_*^n.$$

Proposition 4.2 (Local cscK one-step map). *There are neighbourhoods $U_s \subset H_0^s$ and $V_{s+2} \subset H_0^{s+2}$ of zero such that Zhang's one-step map is a C^∞ map*

$$S_\tau : U_s \longrightarrow V_{s+2}.$$

Its differential at the cscK metric is

$$DS_\tau(0) = (A + \tau B)^{-1} A. \quad (4.5)$$

The same construction gives compatible smooth maps $H_0^m \rightarrow H_0^{m+2}$ for all $m \geq s$.

Proof. Put $\omega_u = \omega_* + \sqrt{-1} \partial \bar{\partial} u$ and define

$$\Phi_\tau(u, v) = R(\omega_v) - \bar{R} - \tau^{-1}(\text{tr}_{\omega_v} \omega_u - n). \quad (4.6)$$

By [\(4.2\)](#), the desired equation is $\Phi_\tau(u, v) = 0$. We first remove a harmless codimension-one issue. Consider

$$\mathcal{F}_\tau(u, v) = P_0 \Phi_\tau(u, v) : H_0^s \times H_0^{s+2} \longrightarrow H_0^{s-2}.$$

Solving $\mathcal{F}_\tau(u, v) = 0$ is equivalent to solving the unprojected equation. To see this, the projected equation says $\Phi_\tau(u, v) = c$ for a constant c . On the other hand,

$$\int_X (R(\omega_v) - \bar{R}) \omega_v^n = 0, \quad (4.7)$$

$$\int_X (\text{tr}_{\omega_v} \omega_u - n) \omega_v^n = n \int_X (\omega_u - \omega_v) \wedge \omega_v^{n-1} = 0. \quad (4.8)$$

The first identity is the cohomological formula for total scalar curvature, and the second follows because ω_u and ω_v are cohomologous. Hence $\int_X \Phi_\tau(u, v) \omega_v^n = 0$. If $\Phi_\tau(u, v) = c$, positivity of the volume gives $c = 0$.

We next verify smoothness. For $v \in H^{s+2}$, the coefficients of ω_v^{-1} lie in H^s and

$$R(\omega_v) = -g_v^{i\bar{j}} \partial_i \partial_{\bar{j}} \log \det(g_{v,p\bar{q}})$$

lies in H^{s-2} . Since $s - 2 > n$, which is half the real dimension of X , all products occurring here are continuous in the indicated Sobolev spaces. Likewise, the second derivatives of $u \in H^s$ lie in H^{s-2} , so $\text{tr}_{\omega_v} \omega_u \in H^{s-2}$. Matrix inversion, determinant, logarithm, differentiation, multiplication, contraction, and P_0 are smooth maps between these spaces. It follows that $\mathcal{F}_\tau$ is C^∞ .

We calculate its derivative. At a cscK metric, the scalar-curvature variation in the direction $\sqrt{-1} \partial \bar{\partial} v$ is

$$\left. \frac{d}{dt} \right|_{t=0} R(\omega_* + t \sqrt{-1} \partial \bar{\partial} v) = -Bv. \quad (4.9)$$

For completeness, differentiating $R = g^{i\bar{j}} R_{i\bar{j}}$ gives

$$\dot{R} = -\Delta_*^2 \dot{v} - \text{Ric}(\omega_*)^{i\bar{j}} \dot{v}_{i\bar{j}},$$

because

$$\dot{g}^{i\bar{j}} = -g_*^{i\bar{p}} \dot{v}_{q\bar{p}} g_*^{q\bar{j}}, \quad \dot{R}_{i\bar{j}} = -\partial_i \partial_{\bar{j}} (\Delta_* \dot{v}).$$

At a cscK metric, the Kähler commutation formula is

$$\mathcal{D}^* \mathcal{D} \phi = \Delta_*^2 \phi + \text{Ric}(\omega_*)^{i\bar{j}} \phi_{i\bar{j}}.$$

Hence the displayed variation is $-\mathcal{D}^* \mathcal{D} \dot{v}$, proving (4.9). The two trace variations are

$$\left. \frac{d}{dt} \right|_{t=0} \text{tr}_{\omega_* + t\sqrt{-1} \partial \bar{\partial} \dot{v}} \omega_* = -\Delta_* \dot{v} = A \dot{v}, \quad (4.10)$$

$$\left. \frac{d}{dt} \right|_{t=0} \text{tr}_{\omega_*} (\omega_* + t\sqrt{-1} \partial \bar{\partial} \dot{u}) = \Delta_* \dot{u} = -A \dot{u}. \quad (4.11)$$

Since $A1 = B1 = 0$, both A and B map zero-mean functions to zero-mean functions. Therefore P_0 does not alter the following linearised operators:

$$D_v \mathcal{F}_\tau(0, 0) \dot{v} = -(B + \tau^{-1} A) \dot{v}, \quad (4.12)$$

$$D_u \mathcal{F}_\tau(0, 0) \dot{u} = \tau^{-1} A \dot{u}. \quad (4.13)$$

The map

$$B + \tau^{-1} A : H_0^{s+2} \longrightarrow H_0^{s-2}$$

is an isomorphism. Indeed, it is a fourth-order self-adjoint elliptic Fredholm operator of index zero. If it annihilates ϕ , then integration by parts and (4.4) give

$$0 = \int_X \phi (B + \tau^{-1} A) \phi \omega_*^n = \|\mathcal{D} \phi\|_{L^2}^2 + \tau^{-1} \|\bar{\partial} \phi\|_{L^2}^2. \quad (4.14)$$

Thus ϕ is constant; its zero mean forces $\phi = 0$. Injectivity and index zero give surjectivity, while elliptic regularity gives a bounded inverse $H_0^{s-2} \rightarrow H_0^{s+2}$.

The Banach implicit-function theorem now produces the smooth local map $v = S_\tau(u)$. Differentiating $\mathcal{F}_\tau(u, S_\tau(u)) = 0$ and using (4.12)–(4.13) yields

$$(B + \tau^{-1} A) \dot{v} = \tau^{-1} A \dot{u},$$

or

$$(A + \tau B) \dot{v} = A \dot{u}.$$

This is (4.5). The same proof at every level $m \geq s$ gives $H_0^m \rightarrow H_0^{m+2}$. The branches agree on overlaps by the uniqueness in Proposition 4.1. $\square$

4.3. The generalised spectrum and an adapted Sobolev norm. The formal expression

$$DS_\tau(0) = (I + \tau C)^{-1}, \quad C = A^{-1} B,$$

is useful, but requires a choice of Hilbert space and operator domain. In this subsection we also use negative Sobolev spaces, with $H_0^t = \{u \in H^t : \langle u, 1 \rangle = 0\}$ for every integer t ; the pairing is taken relative to ω_*^n and is distributional when $t < 0$. Elliptic theory for the Laplacian gives bounded inverses $A^{-1} : H_0^{t-2} \rightarrow H_0^t$.

On the real Hilbert space H_0^1 define

$$(u, v)_A = \langle Av, u \rangle = \text{Re} \int_X \langle \bar{\partial} u, \bar{\partial} v \rangle_{\omega_*} \omega_*^n. \quad (4.15)$$

Poincaré's inequality makes the resulting norm equivalent to the usual H^1 norm. The Laplace spectral resolution shows that A , viewed as an operator on this Hilbert space, is self-adjoint with domain H_0^3 .

Initially define $C = A^{-1}B$ on smooth zero-mean functions. The cscK Lichnerowicz identity gives

$$B = A^2 + R_2, \quad R_2 u = \text{Ric}(\omega_*)^{i\bar{j}} u_{i\bar{j}}, \quad C = A + Q, \quad Q = A^{-1}R_2.$$

Since $R_2 = B - A^2$, it is formally self-adjoint and annihilates constants, so it preserves zero mean. For every integer $m \geq 0$, the maps

$$R_2 : H_0^m \longrightarrow H_0^{m-2}, \quad A^{-1} : H_0^{m-2} \longrightarrow H_0^m$$

are bounded. Consequently Q is bounded on every H_0^m , in particular on the Hilbert space $(H_0^1, (\cdot, \cdot)_A)$.

For smooth zero-mean functions,

$$(Cu, v)_A = \int_X Bu v \omega_*^n = \int_X u Bv \omega_*^n = (u, Cv)_A. \quad (4.16)$$

Subtracting the corresponding identity for A shows that Q is symmetric on smooth functions. Boundedness and density extend this symmetry to all of H_0^1 . The bounded self-adjoint perturbation theorem therefore gives a self-adjoint realization

$$C = A + Q, \quad D(C) = H_0^3.$$

The identity $(Cu, u)_A = \|\mathcal{D}u\|_{L^2}^2 \geq 0$ extends from smooth functions to this domain, so C is nonnegative.

The elliptic estimate for A and boundedness of Q on H_0^1 give

$$\|u\|_{H^3} \leq c(\|Cu\|_{H^1} + \|u\|_{H^1}).$$

Since C is nonnegative and self-adjoint, $(C + I)^{-1}$ exists and is bounded from H_0^1 to H_0^3 . The compact embedding $H_0^3 \hookrightarrow H_0^1$ thus makes this resolvent compact.

Lemma 4.3 (Generalised spectrum). *There is an $(\cdot, \cdot)_A$ -orthonormal basis $\{e_j\}$ of H_0^1 consisting of smooth real functions, and numbers $0 \leq \mu_1 \leq \mu_2 \leq \dots$, repeated with multiplicity and tending to infinity, such that*

$$Be_j = \mu_j Ae_j. \quad (4.17)$$

Moreover,

$$\mu_j = 0 \iff \mathcal{D}e_j = 0 \iff \nabla_{\omega_*}^{1,0} e_j \text{ is holomorphic.}$$

Proof. The compact spectral theorem applied to the positive, injective, self-adjoint operator $(C + I)^{-1}$ gives a complete orthonormal basis with eigenvalues $\nu_j > 0$ tending to zero. Setting $\mu_j = \nu_j^{-1} - 1$ gives nonnegative eigenvalues of C tending to infinity. The equation $Ce_j = \mu_j e_j$ is equivalent to (4.17). Elliptic regularity for the fourth-order operator $B - \mu_j A$ makes every e_j smooth. Finally,

$$\mu_j \|e_j\|_A^2 = (Ce_j, e_j)_A = \|\mathcal{D}e_j\|_{L^2}^2,$$

so $\mu_j = 0$ exactly when $\mathcal{D}e_j = 0$, which is the holomorphic-gradient condition. $\square$

Set

$$K = \ker(B|_{H_0^s}), \quad N = K^{\perp_A} \cap H_0^s. \quad (4.18)$$

The kernel is finite-dimensional. Denote by $\mu_* > 0$ the smallest positive generalised eigenvalue in (4.17). On H_0^s , where $s = 2r + 1$, introduce the graph norm

$$\|u\|_{s,C}^2 = \sum_{\ell=0}^r \|C^\ell u\|_A^2. \quad (4.19)$$

Lemma 4.4 (Equivalence and contraction). *The norm (4.19) is equivalent to the standard H^s norm. The splitting $H_0^s = K \oplus N$ is invariant under $DS_\tau(0)$, and*

$$DS_\tau(0)|_K = \text{Id}, \quad \|DS_\tau(0)|_N\|_{s,C} = \frac{1}{1 + \tau\mu_*} < 1. \quad (4.20)$$

Proof. Since $C = A + Q$ and Q is bounded on every H_0^m , the elliptic estimate for A gives, for every integer $m \geq 0$,

$$\|u\|_{H^{m+2}} \leq c_m(\|Cu\|_{H^m} + \|u\|_{H^m}). \quad (4.21)$$

Moreover, $C : H_0^{m+2} \rightarrow H_0^m$ is bounded. These facts imply $D(C^r) = H_0^{2r+1}$. Indeed, the assertion holds for $r = 1$. Inductively, $u \in D(C^{r+1})$ means $u \in H_0^3$ and $Cu \in H_0^{2r+1}$. The equation $Au = Cu - Qu$, boundedness of Q at every Sobolev level, and elliptic regularity successively improve u to H_0^{2r+3} . The reverse inclusion follows from boundedness of $C : H_0^{m+2} \rightarrow H_0^m$.

Each term $\|C^\ell u\|_A$ is bounded by a constant times $\|u\|_{H^{2\ell+1}}$, proving $\|u\|_{s,C} \leq c\|u\|_{H^s}$. Conversely, starting with the H^1 norm, which is equivalent to $\|\cdot\|_A$ on zero-mean functions, iterate (4.21). At the ℓ -th step all terms that arise are bounded by $\|C^j u\|_A$, $0 \leq j \leq \ell$. After r steps this yields

$$\|u\|_{H^{2r+1}} \leq c \sum_{\ell=0}^r \|C^\ell u\|_A \leq c\sqrt{r+1} \|u\|_{s,C}.$$

Hence the two norms are equivalent. Since $H_0^s = D(C^r)$, the spectral expansion converges in this graph norm.

If e_j satisfies (4.17), then (4.5) gives

$$DS_\tau(0)e_j = \frac{1}{1 + \tau\mu_j} e_j. \quad (4.22)$$

It follows immediately that K and N are invariant, that the restriction to K is the identity, and that all multipliers on N are at most $(1 + \tau\mu_*)^{-1}$. Expanding in the A -orthonormal eigenbasis, every term in the graph norm receives the same multiplier. This proves the operator-norm identity in (4.20). $\square$

4.4. The local cscK fixed manifold.

Proposition 4.5. *In an H^s neighbourhood of ω_* , the $\text{Aut}_0(X)$ -orbit contains a finite-dimensional smooth embedded submanifold $\mathcal{M}_{\text{cscK}}$ of cscK metrics fixed by S_τ , and*

$$T_{\omega_*} \mathcal{M}_{\text{cscK}} = K = \ker(B|_{H_0^s}).$$

Proof. By (4.4), K consists precisely of the zero-mean real functions ϕ for which $\nabla_{\omega_*}^{1,0}\phi$ is holomorphic. For such a ϕ , let W_ϕ be the corresponding real holomorphic gradient vector field, scaled so that

$$\iota_{W_\phi} \omega_* = d^c \phi \quad \text{and hence} \quad \mathcal{L}_{W_\phi} \omega_* = dd^c \phi = \sqrt{-1} \partial \bar{\partial} \phi$$

with our fixed convention for d^c . Let f_ϕ be its time-one flow. The flow is holomorphic and depends smoothly on the finite-dimensional parameter ϕ . Since it is isotopic to the identity, it acts trivially on cohomology. Thus, for ϕ sufficiently small, the $\partial\bar{\partial}$ -lemma and the zero-mean normalisation give a unique potential $p(\phi) \in H_0^s$ such that

$$f_\phi^* \omega_* = \omega_* + \sqrt{-1} \partial \bar{\partial} p(\phi)$$

and the elliptic solution operator for $\sqrt{-1} \partial \bar{\partial}$ shows that p is smooth. Differentiating at the origin yields

$$\sqrt{-1} \partial \bar{\partial} (Dp(0)\phi) = \left. \frac{d}{dt} \right|_{t=0} f_{t\phi}^* \omega_* = \mathcal{L}_{W_\phi} \omega_* = \sqrt{-1} \partial \bar{\partial} \phi.$$

Consequently $Dp(0)\phi - \phi$ is constant, and its zero mean forces $Dp(0)\phi = \phi$. The Banach constant-rank theorem, applied to the finite-dimensional domain K , now shows that the local image of $p : K \rightarrow H_0^s$ is an embedded finite-dimensional submanifold tangent to K .

Holomorphic pullback preserves scalar curvature:

$$R(f_\phi^* \omega_*) = f_\phi^* R(\omega_*) = \bar{R}.$$

Thus every metric in the image is cscK. By [Proposition 4.1](#), every such metric is a fixed point of S_τ . We take the local image to be $\mathcal{M}_{\text{cscK}}$. Globally, the fact that any two cscK metrics in one Kähler class differ by an automorphism is the uniqueness theorem of Berman–Berndtsson [2]; only the elementary local orbit construction is used here. $\square$

4.5. Proof of local attraction and the global corollary.

Proof of Theorem 1.2. Regard the local map from [Proposition 4.2](#) as a C^1 self-map of H_0^s . By [Proposition 4.5](#), it fixes the finite-dimensional submanifold $\mathcal{M}_{\text{cscK}}$ pointwise. Its tangent space at the origin is K . The A -orthogonal complement N in (4.18) is invariant, and [Lemma 4.4](#) gives strict operator-norm contraction on N in a norm equivalent to H^s . All assumptions of [Proposition 2.1](#) are therefore satisfied. We obtain geometric H^s convergence to a point of $\mathcal{M}_{\text{cscK}}$.

The limiting metric ω_∞ is smooth and cscK by the construction of $\mathcal{M}_{\text{cscK}}$. As in the proof of [Theorem 1.1](#), use ω_∞ as the reference metric and subtract the ω_∞^n -mean from $u_k - u_\infty$ to obtain normalized potentials $w_k \rightarrow 0$ in H^s . Reapply [Proposition 4.2](#) with this reference metric. For every $m \geq s$, the resulting local map near zero gains two derivatives and represents the same iteration by uniqueness. The Sobolev-scale bootstrap [Lemma 2.5](#), now centered at the actual limit, gives convergence in all Sobolev norms, hence in C^∞ . Taking $\mathcal{U}_{\text{cscK}}$ to be the resulting local basin proves [Theorem 1.2](#). $\square$

Proof of Corollary 1.5. Zhang’s modulo-automorphism theorem [24, Theorem 6.6] supplies automorphisms $g_k \in \text{Aut}_0(X)$ and a cscK metric ω_* such that

$$g_k^* \omega_k \longrightarrow \omega_* \quad \text{in } C^\infty.$$

Choose one sufficiently large N so that $g_N^* \omega_N$ belongs to the local basin $\mathcal{U}_{\text{cscK}}$ of [Theorem 1.2](#), and freeze the single automorphism $g = g_N$. By (4.3),

$$g^* \omega_{N+j} = S_\tau^j(g^* \omega_N), \quad j \geq 0.$$

The right-hand side converges smoothly to a cscK metric η_∞ . Pulling back by the one fixed inverse gives

$$\omega_{N+j} \longrightarrow (g^{-1})^* \eta_\infty \quad \text{in } C^\infty.$$

The limit is cscK, and adding back the finite initial segment does not affect convergence. This removes the varying sequence of automorphisms from Zhang’s theorem. $\square$

5. CONSEQUENCES, LIMITATIONS, AND CONTEXT

5.1. The limit is not a prescribed canonical metric. The local theorems [Theorems 1.1](#) and [1.2](#), and consequently the global corollaries [Corollaries 1.4](#) and [1.5](#), assert convergence to *some* canonical metric; they do not select a predetermined point on the automorphism orbit. This distinction is necessary. If ω_* is canonical and $h \in \text{Aut}_0(X)$, then $h^* \omega_*$ is again canonical and is a fixed point of the relevant iteration. Starting at

$$\omega_0 = h^* \omega_*$$

therefore gives the stationary sequence $\omega_k = h^* \omega_*$ for every k . Unless h is an isometry of ω_* , this limit differs from ω_* . The summable tangential drift in (1.4) determines which point of the fixed manifold is selected.

5.2. Why a finite time step is essential. For every finite τ , the positive twisting term in each one-step equation gives uniqueness and hence a single-valued equivariant map. Formally sending τ to infinity in the Fano equation leaves

$$\text{Ric}(\beta) = \beta,$$

which no longer depends on the input and generally has an entire automorphism orbit of solutions. A choice of a different solution at every step could jump indefinitely along that

orbit. Thus our conclusion concerns each fixed $\tau \in (0, \infty)$; it is not a statement about an arbitrary multi-valued “time-infinity” correspondence.

5.3. Rates and the role of the old gauges. In the Fano case, the derivative of the normal block has exact spectral norm

$$q_{\text{Fano}} = \frac{1}{1 + \tau(\lambda_* - 1)},$$

where $\lambda_* > 1$ is the first Laplace eigenvalue above the holomorphy potentials. In the cscK case it is

$$q_{\text{cscK}} = \frac{1}{1 + \tau\mu_*},$$

where $Be = \mu Ae$ and $\mu_* > 0$ is the first positive generalised eigenvalue. By continuity of the nonlinear derivative, the local orbit converges at every geometric rate slightly larger than the corresponding number.

No assertion about convergence of Zhang’s or Darvas–Rubinstein’s original gauge sequence $\{g_k\}$ is needed. Those gauges serve only to guarantee that one late gauged iterate enters the local basin. The proof then discards the entire sequence and retains the single map g_N . This is precisely the content of the gauge-freezing principle [Proposition 2.4](#).

5.4. Relation with earlier work. The inverse Ricci and Ricci iterations originated in [\[21, 22\]](#) and were further studied in Kähler classes by Keller [\[15\]](#), through Monge–Ampère iteration by Hunter [\[14\]](#), and in homogeneous Riemannian settings in [\[20, 6\]](#). Darvas–Rubinstein’s global Fano convergence theorem combines variational methods on finite-energy spaces with properness modulo automorphisms [\[11, 12, 13\]](#). Zhang’s extension to arbitrary Kähler classes relies on the cscK estimates and existence theory of Chen–Cheng [\[8, 9\]](#), the metric geometry and regularity of the extended K-energy [\[3, 4\]](#), and the uniqueness theory of [\[2\]](#).

The local mechanism used here is the discrete analogue of stability results for geometric flows. For the pseudo-Calabi flow, local convergence was proved by Chen–Zheng [\[10\]](#); for the Kähler–Ricci flow, spectral and automorphism effects have been studied, among others, in [\[19, 18\]](#). Our argument does not require a comparison between the discrete iteration and its continuous flow. It uses only the elliptic one-step equation, the canonical fixed manifold, and the already established global convergence modulo automorphisms.

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